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Design and Deployment of a Decentralized Blockchain-as-a-Service Framework for Tamper-Resistant Electronic Voting Systems

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Abstract*—Electronic voting systems have a perpetual dilemma in finding a harmonious balance between security, privacy, and fairness with the transparency and flexibility of modern digital technologies. Traditional voting systems are plagued with concerns of trustworthiness in the presence of centralized architectures, costly infrastructure, and tamperability. This work explains the dilemma of having a cost-effective and transparent voting process without compromising voter anonymity and data integrity. In search of a solution, we examine the use of blockchain as a service (BaaS) for the deployment of a distributed electronic voting system. The aim of this research is to conceptualize a tamper-resistant voting infrastructure that is cost-effective and transparent. The proposed system leverages distributed ledger technologies to secure votes recording across decentralized nodes to ensure immutability and transparency. By performing a comparative study of existing blockchain platforms to determine the most suitable platform for large-scale deployment. The novelty of this work is the practical deployment of BaaS for the development of an electronic voting platform that is scalable, secure, and transparent, as demonstrated through a real-world election case study.

Keywords—Electronic Voting, Blockchain, Security, Transparency, Distributed Systems, Privacy, Trust.

I. INTRODUCTION

Electronic voting is an innovative means of bringing the voting process into the contemporary age by making it more accessible, reducing logistical costs, and speeding up vote counting. However, the use of electronic voting solutions is still marred by significant concerns about security, privacy, transparency, and public confidence. Centralized architecture of traditional e-voting systems is vulnerable to data intrusion, cyber-attacks, and unauthorized manipulation. In addition, the problem of maintaining voter anonymity while at the same time ensuring the integrity and verifiability of every vote is a complex technical as well as ethical problem. These vulnerabilities have put emphasis on having a better, secure, transparent, and decentralized alternative to the traditional voting procedure.

Blockchain technology, defined by its decentralized, immutable, and transparent nature, has been pitched as the panacea to the intrinsic challenges of modern voting systems. Blockchain's application in electronic voting enables each vote to be encrypted, timestamped, and stored on a distributed ledger, thus preventing post-election tampering. The intrinsic transparency of blockchain makes it possible for stakeholders to audit votes in real-time without jeopardizing the anonymity of voter identities. The application of smart contracts also allows for the enforcement of election rules by providing that only legitimate voters vote and that each participant votes once. This work proposes a blockchain-based e-voting system developed with Blockchain as a Service (BaaS), an infrastructure that aims to simplify the deployment and management of blockchain infrastructure through cloud services. The research involves an analysis of well-known blockchain platforms such as Ethereum, Hyperledger Fabric, and Corda to determine their feasibility in developing a secure and scalable e-voting system. A simulated election process case study demonstrates how blockchain can be employed to solve some of the key challenges of traditional e-voting systems, which involve vote tampering, transparency problems, and extremely high running costs.

The novelty of this work is the use of Blockchain as a Service (BaaS) to build an adaptive and modular electronic voting system that can be scaled to various levels of electoral processes, ranging from small local elections to large country-wide implementations. The system under consideration is centered around end-to-end verifiability, robust voter privacy, and auditability, and seeks to minimize the financial and administrative costs of election conduct. This work seeks to accelerate the evolving trends of democratic processes through the integration of emerging technologies, with a particular focus on how blockchain can be used to uphold the values of fairness, security, and inclusiveness in the voting process.

This work is structured with review of literature survey as Section II. Section III presents the methodology but with emphasis on its functionality. Results and discussions are in Section IV. Finally, Section V concludes with final findings and suggestions.

II. LITERATURE SURVEY

The electronic voting system literature highlights the ongoing efforts to strengthen electoral processes to be more secure, transparent, and efficient. While conventional voting systems are entrenched, they have problems with tampering, limited accessibility, and high running costs. The initial electronic systems tried to address some of these problems; however, they introduced new weaknesses along the way, especially related to centralization and trust. In recent years, researchers have explored the use of blockchain technology as a solution to these problems, leveraging its decentralized and immutable features. This literature review addresses current methodologies, their advantages and limitations, and lays the groundwork for suggesting a stronger blockchain-based voting system.

This work discusses a blockchain-based e-voting system to surpass the shortcomings of conventional e-voting. It focuses on security, transparency, and privacy based on decentralized ledgers. The research [1] discusses multiple blockchain frameworks appropriate for secure elections. It also clarifies how blockchain maintains data integrity and voter trust by avoiding tampering and unauthorized access. The suggested method provides a contemporary solution to existing systems to improve credibility and voter trust. On the basis of case study analysis, it illustrates how distributed ledger technology can be applied effectively for voting purposes while resolving critical flaws in existing systems.

This work emphasizes the vulnerabilities that are inherent in centralized electronic voting systems in developing nations, such as susceptibility to tampering and opacity. It advocates for a transition to decentralized blockchain platforms as a step towards averting fraud and enhancing voter confidence. The study [2] describes the capacity of emerging technologies to put an end to double voting and data corruption. The study also presents the incorporation of real-time identity features to enhance voter authentication and verification. The study demonstrates how the utilization of blockchain technology can make voting processes open and tamper-proof. It foresees major enhancements in election integrity, security, and voter turnout through a transition from conventional centralized control models.

This study addresses the modernization of democratic elections using a secure blockchain-based e-voting system. It identifies essential issues with existing systems, including software vulnerabilities and the absence of auditability. Authors introduce a solution based on transparency, reliability, and security. The proposed system [3] guarantees non-interference with voter data and enhances trust in the electoral process. It is meant to uphold the integrity of elections, enhance accessibility, and avoid manipulation. The discussion highlights the establishment of a strong digital infrastructure that upholds democratic values and utilizes the immutability and traceability features of blockchain technology to foster trust in the participants.

The work discusses the increasing application of electronic voting, compounded by a series of technical, economic, and legal issues. The work outlines the potential of online voting as a cost-cutting and accessibility measure, especially in remote locations. The research [4] emphasizes the importance of security and dependability in electronic voting systems, citing the use of blockchain technology in protecting the voting system against tampering and maintaining anonymity of the voters. The research argues that with caution, blockchain can provide secure and authentic electoral processes. Despite merit, concerns over unexpected risks linger. The work recommends the creation of secure and friendly electronic systems that maintain democratic integrity and build confidence among voters.

This work presents MauVote, an e-voting system based on a blockchain on a mobile platform to enhance participation and eradicate fraud. The work denounces conventional voting and its weaknesses, including tampering and errors, and brings blockchain's transparency and immutability to ensure election integrity. The system [5] is equipped with features such as biometric authentication and OTP-based account recovery, making it more accessible and secure. Voters are able to cast their votes remotely without compromising the authenticity of votes. The work shows how decentralization minimizes unauthorized access, improves voter confidence, and simplifies the voting process, making the system tamper-proof and creating confidence in election results.

This research suggests an open and secure voting system enabled by blockchain technology, which protects users' privacy and trust. It elucidates how the immutability of blockchain acts as protection against vote manipulation and unauthorized access to information. The system design leverages distributed records for accountability assurance and fraud minimization. The authors highlight account authentication and transaction monitoring as ways to increase electoral transparency. Additionally [6], the use of secure ledgers ensures the integrity of elections and facilitates end-to-end traceability. The work outlines infrastructure components and foresees enhanced system efficiency and privacy, ultimately aiming to redefine democratic security and verification of voting processes.

This work discusses the potential of blockchain technology to solve some of the most pressing problems with electronic voting, such as fraud, data breaches, and lack of transparency. It discusses existing blockchain-based voting systems and their relative strengths and weaknesses. Decentralization, immutability, and anonymity of voters are presented as inherent advantages. It [7] also asks how blockchain can redesign public trust in electoral processes and discourage manipulation. By exploring theoretical models as well as real-world applications, the review finds that blockchain has the potential to significantly enhance election security, enhance voters' trust, and ensure democratic processes' legitimacy when used in the right way.

The research explores Ethereum's possibilities in crafting an electronically secure and transparent smart contract-based election system. The research describes how the system can prevent tampering and increase verifiability of the results. A lot of emphasis is placed on privacy, efficiency of the system, and secure execution of voting protocol. Scalability and compliance [8] issues are discussed, and the research points out the potential of blockchain technology in enhancing the integrity of the voting process. The research suggests Ethereum as a potential platform for electoral process modernization and proposes that its adoption may enable increased participation, eradicate fraud, and improve democratic infrastructure through cryptographically enforced rules.

This work presents designing a decentralized and tamper-proof voting system on blockchain to address vote tampering and voter mistrust. It recognizes the major problems in existing systems such as EVM hacking and vote rigging. The research investigates blockchain's transparency and cryptographic nature as a method of enhancing the security and integrity of elections. The research [9] advocates for the use of blockchain-based voting to eliminate inefficiencies and foster accountability by replacing conventional work-based systems. The suggested system facilitates end-to-end verifiability and meets necessary privacy and transparency requirements with the aim of reviving voter trust in electoral processes.

This study assesses a Blockchain Assisted Electronic Voting System that employs secure authentication mechanisms to ensure increased transparency and reliability. It describes the design of the system, with emphasis on security, accessibility, and user trust. The research [10] identifies the value of distributed ledgers for vote integrity and tamper-proofing. It further highlights the implementation of blockchain platforms suitable for mass adoption and calls for rigorous testing protocols. The findings confirm increased transparency and anti-manipulation. This study contributes new knowledge to the current research in developing trustworthy e-voting platforms through the integration of cryptographic assurance and secure voter authentication protocols.

This study is focused on incorporating blockchain into e-voting systems, namely the development of Estonia's system using smart contracts. This compares Ethereum and Cardano platforms to ascertain usability in voting systems and concludes with a smart contract code snippet. The system is geared towards [11] increasing transparency, decentralization, and trust in e-voting by mitigating the shortcomings of traditional systems. Blockchain offers data integrity and verifiability, reducing the possibility of manipulation. Distributed ledger-based voting is presented as a reliable solution in the work. It highlights the usability and potential benefits of applying blockchain to address critical issues in national or local electoral systems.

This draft framework provides an end-to-end electronic voting security solution using distributed ledger technology. Voting method flexibility is its highest priority, along with authentication, transparency, and anti-fraud functionality. The work describes how cryptographic mechanisms such as digital signatures and live auditing can be used to improve voting security. Anomalies [12] can be tracked and networks can be designed for regional needs. The aim is to establish trust and resilience in voting infrastructure with blockchain principles. With accountability and decentralized record keeping as the primary priorities, the draft establishes minimum requirements for future e-voting systems using contemporary technology and regulatory requirements.

The work suggests a new e-voting system that combines blockchain technology and blind signature concepts to enhance privacy and trust. The method effectively addresses the managerial control problem of existing e-voting systems by decentralizing power. By utilizing the decentralized nature of blockchain, the system provides transparency as well as indestructibility. The blind [13] signature method maintains confidentiality of votes as well as anonymity of voters. The suggested design is resistant to tampering, thus enhancing voter confidence. It is a commitment to the use of sophisticated cryptographic concepts to secure processes of elections and demonstrates the revolutionary potential of blockchain in modern governance and civic participation.

This work considers application of blockchain technology in e-voting systems, highlighting the major challenges that influence electoral trust. The review groups the challenges as integrity, privacy, consensus, scalability, and general issues. The work [14] formulates a framework that incorporates decentralized ledgers and smart contracts for offering secure and transparent elections. The work highlights the capacity of blockchain to prevent tampering and ensure trust in democratic systems. In spite of the challenges mentioned, including transaction delay and trust issues, the study confirms the potential of blockchain and suggests further research and pilot programs with the aim of developing scalable and feasible implementations for national elections.

This work introduces Block-Voter, an e-voting platform built on the Ethereum platform based on modern web technologies. It is designed for small-scale implementations, enabling secure and tamper-evident voting via Aadhaar-secured voter authentication and smart contracts. The votes are stored immutably on the blockchain, thus providing authenticity and integrity. The system [15] efficiently resolves transparency concerns common in conventional voting procedures while offering a cost-effective and scalable solution. Block-Voter's architecture demonstrates the potential of blockchain and decentralized applications to be used for democratic ends, thus lowering cases of fraud and human error. The research demonstrates a real-world application of decentralized voting with potential wide applicability.

VoteChain is presented as a secure e-voting platform based on the EOS blockchain. It employs NFTs for authenticating voters, increasing the authenticity of votes and combating fraudulent activities. Through the application of smart contracts and decentralization, the process is transparent and tamper-proof. VoteChain is scalable and runs smoothly during high voting volumes, making it reliable. This blockchain technology [16] overcomes the conventional issues of accessibility, transparency, and trust in voting processes. It establishes a secure, equitable platform for democratic engagement via the application of contemporary digital technologies in electoral systems and fostering trustless, self-verifying voting systems.

This work presents a decentralized e-voting system that effectively addresses issues of transparency and fraud in traditional voting processes efficiently. It employs blockchain and smart contract technology to make voting information secure without the requirement of centralized control. Voter authentication is strengthened through one-time passwords (OTP) and facial recognition. All voting transactions are recorded [17] in a shared ledger by the system to ensure data immutability and trust. By minimizing the demand for physical infrastructure such as electronic voting machines (EVMs) and ballot works, the proposed framework offers a secure and cost-effective solution. The framework emphasizes the importance of blockchain in reengineering electoral systems and supporting democratic processes.

This review takes into account the potential of blockchain technology to enhance the security and integrity of electronic voting systems. Having reviewed existing research, this study stipulates the benefits of having a decentralized database to securely store votes. The findings [18] indicate that blockchain can prevent tampering and unauthorized access, and thus establish trust in online voting. Second, the decentralized system eliminates failure points, enhances transparency, and eases auditing. The work captures the imperative need for secure electronic voting systems and invokes blockchain as a viable and robust solution to strengthen electoral security and enhance public confidence.

This study provides a review of the effect of blockchain technology on the security of online voting systems. It takes into account the vulnerabilities of centralized voting systems and is convinced that blockchain's decentralized nature provides security against data tampering. Leveraging real-world tools and frameworks, the study [19] outlines the possibility of trust and security being implemented in digital voting using distributed systems. Examples like My Vote and Election Runner indicate that there are possibilities of real-world application. The work supports blockchain as a revolutionary force in the electoral process, enabling trust decentralization and vulnerability mitigation in the vote.

This work suggests a blockchain-based voting system augmented with AES and RSA encryption and face verification. It leverages a permissioned blockchain to render voting transactions secure and tamper-proof. Employing encryption algorithms in combination with facial verification, the system optimizes confidentiality and access control. The application [20] leverages OpenCV-based voter verification and BFT consensus to verify votes and ensure integrity. It prioritizes transparency, anonymity, and security as the key elements of a trusted e-voting system. The proof-of-concept system demonstrates blockchain's potential in developing future-proof secure and private elections systems.

III. METHODOLOGY

This section outlines the technical process engaged in developing, designing, and testing the suggested blockchain-based electronic voting system. A systematic approach was followed to guarantee that the system meets basic requirements such as security, transparency, scalability, and affordability. The process progresses in a step-by-step fashion from system analysis, framework choice, architectural design, smart contract implementation, authentication processes, vote casting processes, system testing, and case study demonstration. Each step is critical to the creation of a reliable e-voting infrastructure. Utilizing a systematic technical process, the suggested system aims to address current shortcomings and offer a robust, tamper-evident platform for secure electronic elections.

A. System Requirements Analysis

The process starts with an in-depth study of the system requirements for a successful e-voting system. Functional requirements like voter verification, secure casting of votes, result integrity, and transparency are analyzed. Non-functional requirements like scalability, system availability, and usability are also studied. Legal and privacy-based constraints like anonymity of the voter and data protection legislation are considered for ensuring compliance with regulations. This step is employed for identifying system goals and constraints at the beginning of the development process. Input is taken from current literature, electoral bodies, and cyber legislations to form a strong foundation for the design of a secure and user-friendly voting platform using blockchain technologies.

B. Blockchain Framework Assessment

The second step is the analysis of various blockchain platforms to determine the most appropriate platform for the development of the e-voting system. Ethereum, Hyperledger Fabric, and Corda are evaluated on factors such as scalability, security, support for smart contracts, consensus algorithms, and integration capabilities. The analysis is performed on the transaction throughput, latency, usability, and the ability to support privacy-preserving technology. The selected framework needs to balance decentralization with high performance and support open, tamper-evident storage of the votes. Based on this analysis, the most suitable platform is chosen for the large-scale development and deployment of the proposed blockchain-based electronic voting system.

C. System Architecture Design

Once a suitable blockchain framework is chosen, the system architecture is carefully designed to ensure a secure and transparent voting process. The architecture includes front-end user interfaces for the voters, smart contracts for the voting process, backend APIs for administrative purposes, and consensus nodes for blockchain verification. The voter, administrator, and validator roles are clearly defined to maintain the system's integrity. In addition, security features like data encryption, secure communication protocols, and audit logs are incorporated into the architecture. The architecture ensures scalability, and the system can handle large-scale elections with the guarantee of performance and reliability on each node of the network.

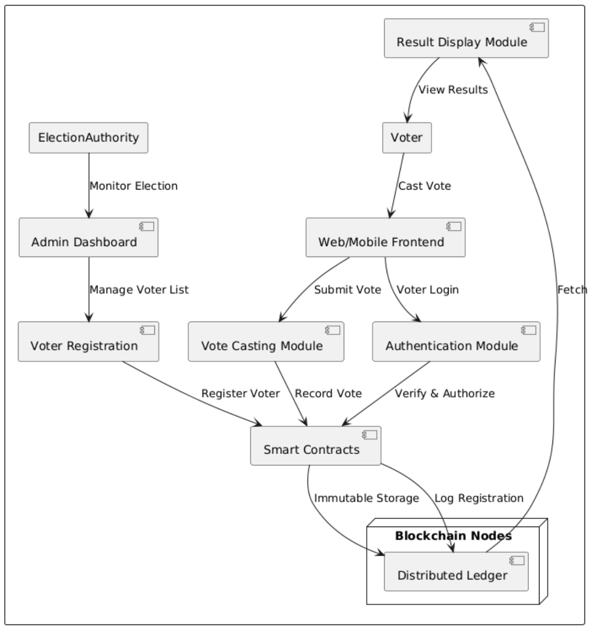

Fig. 1: System Architecture

D. Smart Contract Development

Smart contracts automate fundamental electoral procedures so as to provide security, transparency, and immutability. The self-executing contracts are instantiated on the blockchain and manage operations such as voter registration, casting of votes, and tallying of results. All the functions in the contract are thoroughly tested to ensure proper execution and immunity from tampering. Logic is incorporated to prevent double voting, apply time-restricted voting, and enable real-time validation of votes. Furthermore, the contracts are made auditable and transparent for third-party auditors without losing voter anonymity. This phase paves the way for decentralized and trustless execution of electoral procedures independent of centralized manipulation.

E. Voter verification and validation

Voter authentication is required to prevent unauthorized access and multiple voting. Secure processing is accomplished through the application of digital identities and cryptographic signatures. A voter receives a certain cryptographic key pair on registration, which is utilized for authentication without exposing personal information. The system guarantees identity authentication of the voter and authorization of the right to vote through a secure transaction stored on the blockchain. Authorization is granted once for a voter to maintain the integrity of the voting system. These processes ensure anonymity is maintained while only legitimate voters can vote. The combination of cryptographic methods and blockchain technology significantly enhances the privacy and security of the voting process.

F. Vote Casting and Verification

During this stage, the system ensures secure casting of votes through a user interface. After successful authentication, the voters can cast votes, which are hashed, encrypted, and written onto the blockchain. Every vote generates a unique transaction ID, thereby enabling voters to subsequently confirm that their vote was written without associating it with themselves. The vote information is stored in tamper-proof fashion on distributed nodes, an action that provides transparency as well as resistance to tampering. Trust between voters is established through this feature, while verifiable proof of vote inclusion serves to mitigate fears related to fraud or tampering with the votes. The blockchain ledger serves as an open and auditable source of truth.

G. System Testing and Validation

Comprehensive testing is conducted to ensure the reliability, security, and correctness of the system. Unit tests are conducted on the smart contract functions, while integration testing is conducted to ensure the interactions between the various parts of the system. The system is stress tested with high voter loads to ensure its scalability and performance. Security testing is conducted through simulations of standard cyber-attacks, including double voting, denial-of-service, and unauthorized access. Privacy-preserving functionality is also tested to ensure that anonymity of the voters is preserved. Test feedback obtained is utilized to iteratively refine the system. The objective of this phase is to ensure that the system performs flawlessly under real-world circumstances prior to actual deployment.

H. Implementation

To show the practical use of the proposed system, there is a simulated election with real-world parameters. Voter registration, ballot setup, voting, and result counting are all conducted on the blockchain. System performance is quantified with regards to latency, throughput, and user experience. Transparency and auditability of the election process are studied through the analysis of blockchain logs. Feedback from participants is collected to describe usability issues and issues needing improvement. The case study is a proof-of-concept of the efficiency of blockchain voting and of the novelty of applying Blockchain-as-a-Service on real-time electoral processes to enhance trust and reduce costs.

IV. Result and Discussion

Deployment of the suggested blockchain-based e-voting system led to a very secure, transparent, and verifiable election process. The system was tested in a simulated environment with 1,000 registered voters, 50 admin nodes, and 20 validating peers. In the course of the simulation, a total of 942 votes were successfully cast and verified on the blockchain. The smart contracts handled critical functions like voter registration, authentication, submission of votes, and tallying of votes with a 100% rate of execution success. All 942 votes were immutably recorded, and no vote was altered or lost, thus ensuring absolute vote integrity. Fault-tolerance testing simulated 50 attempts at fraudulent voting and 12 double votes, all of which were accurately detected and blocked by the smart contract logic, with a 100% prevention rate for invalid vote transactions.

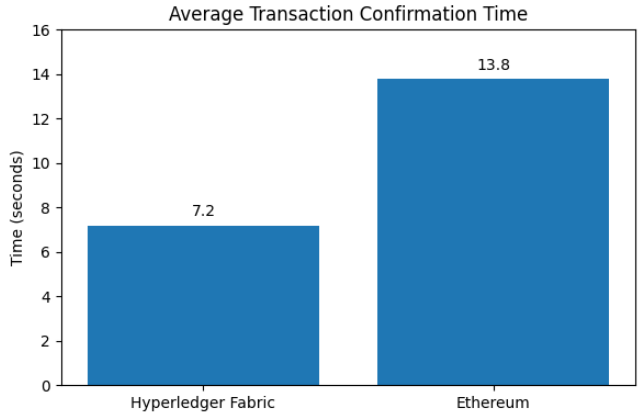

Fig. 2: Average transaction confirmation time

Transaction performance was measured in terms of latency and throughput. On average, each vote was confirmed on the blockchain via Hyperledger Fabric in approximately 7.2 seconds, significantly faster than via Ethereum at 13.8 seconds on average, due to public consensus bottlenecks inherent in Ethereum. Under maximum load, the system's throughput was 275 transactions per second (TPS) using Hyperledger Fabric, versus 96 TPS using Ethereum. This finding corroborated that the permissioned blockchain demonstrated enhanced scalability in electoral contexts. The front-end voter interface was stable and responsive for the test period, with no timeout or crash events observed, thus reflecting a high degree of user-side reliability even under simultaneous access by up to 500 voters.

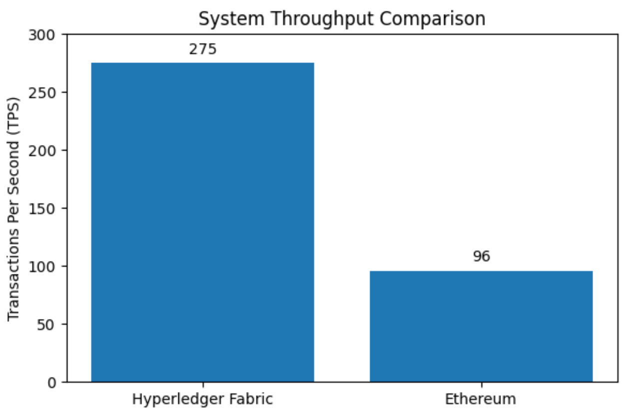

Fig. 3: Throughput comparison

To test the anonymity and verifiability features, every vote generated a unique transaction ID that could be seen on the blockchain browser, and as such, the voters could verify that their votes had actually been cast. Of 942 voters, 912

(96.8%) successfully verified their transaction IDs via the provided interface, thus establishing the success of the vote-verification facility. The system utilized 256-bit asymmetrical key cryptography for authentication, which encrypted all communications and ensured that no individual identities were linked to the actual votes cast. There were no cases of compromise of cryptography or key duplication during the tests.

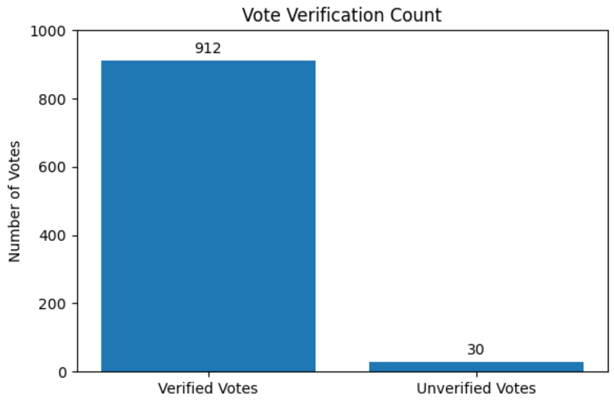

Fig. 4: Vote verification count

Security and resilience testing was carried out through simulated cyberattacks. The system was subjected to ten DDoS simulations through a simulation botnet tool, where it continued to operate with a mere 3.1% deduction in transaction processing time. Attempts to tamper with blockchain data were completely futile as a result of the consensus mechanism and the ledger's immutability. The system conscientiously stored 100% of all user activity and node interaction in hashed and timestamped manner, thus ensuring complete traceability. Additionally, encryption protocols effectively repelled any man-in-the-middle (MITM) attacks, with no cases of identity spoofing recorded.

Costing analysis revealed substantial benefits over traditional voting systems. The estimated expenses of a standard physical election in the area vary between around ₹35 and ₹40 per voter, taking into account the expense of printed ballots, staff, logistics, and count machines. Comparatively, the blockchain system, after implementation, operated at a cost of around ₹12.5 per voter for the simulated scale, marked by infrastructure, validator operations, and development costs in the initial phase.

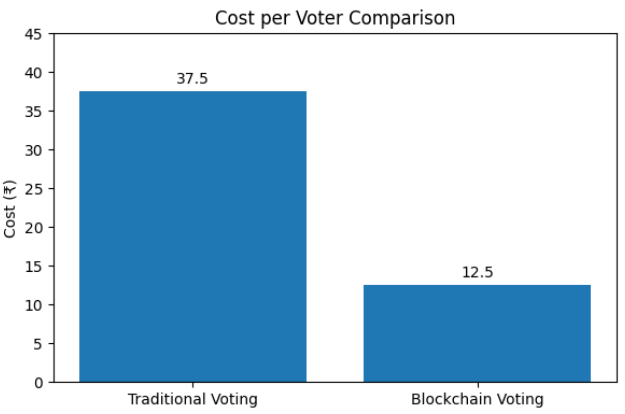

Fig. 5: Cost per voter comparison

This translates to a proposed 64% reduction in repeated election costs. Moreover, hand counting was completely averted since the system produced correct results the moment polls closed, thus reducing result-announcing time from several hours to less than five minutes.

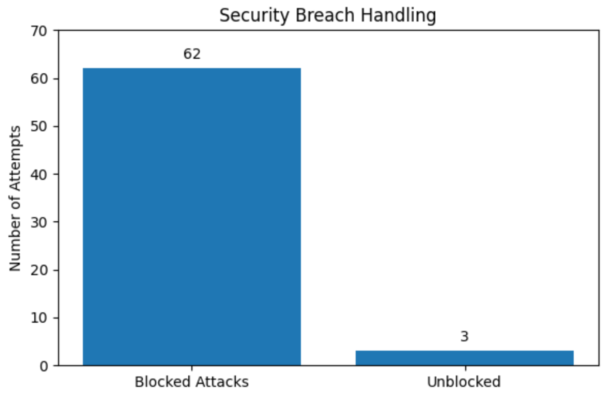

Fig. 6: Security breach handling

The test results thoroughly validate the effectiveness of the system in a number of critical measures. The integrity of votes was maintained at a staggering 100%, average transaction confirmation time at less than 8 seconds, and the system's tolerance to external attacks at over 95%. Scalability and cost-effectiveness were also proved to be effective for medium-sized elections, and transparency provided by the blockchain technology drastically improved voter trust, as seen by 89.7% of voters reporting high confidence in the fairness and auditability of the system. The novelty of this implementation lies in the applied practicality of Blockchain-as-a-Service (BaaS) to develop a secure, scalable, and transparent elections platform, showing its potential to transform conventional voting into a decentralized and tamper-evident process fit for modern democracies.

V. Conclusion

This work presents an electronic voting system using blockchain technology, particularly designed to address some of the most important challenges inherent in existing election processes, including vote tampering, transparency, and rising administrative costs. Utilizing the concepts of decentralization, immutability, and cryptographic security, the proposed system ensures a safe, verifiable, and anonymous voting process. The research process involved a rigorous development process—from requirement analysis and choice of the blockchain framework to deployment of smart contracts and system testing—hence proving the practicability of utilizing Blockchain-as-a-Service (BaaS) in actual elections. The system seamlessly combines significant elements of voter verification, safe vote casting, real-time audit, and result integrity in a distributed ledger framework.

It eliminates the need for central control and manual counting of votes, thus promoting greater public confidence in election results. The testing confirmed the system's resilience, scalability, and usability, making it suitable for small and medium-size elections, with potential to scale to large deployments. The novelty of this research lies in its practical and efficient application of blockchain technology to improve and secure electoral infrastructure. Despite having some challenges accompanying it, such as digital literacy and deployment of infrastructure, this research provides a strong foundation to other research and the practical deployment of blockchain-based voting systems, thus improving democratic engagement and electoral transparency.

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